12/11/2025

Code-B_(Phase-2)

Aakash
Medical | IIT-JEE | Foundations

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MM : 720 NCERT Booster Test Series for NEET-2026_RM(P2)_NBTS-02B **Time : 180 Min.**

Topics Covered:
Physics: Gravitation, Mechanical Properties of Solids, Mechanical Properties of Fluids, Thermal Properties of Matter, Thermodynamics, Kinetic Theory, Oscillations, Waves, Electric Charges & Fields
Chemistry: Thermodynamics, Equilibrium, Redox Reactions, Electrochemistry, Solutions
Botany: Morphology of Flowering Plants, Anatomy of Flowering Plants, Plant Kingdom
Zoology: Excretory Products & their Elimination, Locomotion & Movement, Neural Control & Coordination, Chemical Coordination & Integration

General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

Vivek Krishna

PHYSICS

1. Kepler's 2nd law (law of areas) can be considered as a consequence of which of the following laws?

(1) Conservation of energy

(2) Conservation of linear momentum

(3) Conservation of angular momentum

(4) Conservation of mass

2. A satellite is orbiting just above the surface of earth with period T. If ρ is the density of earth and G is universal gravitational constant, then the quantity $\frac{G\rho}{3\pi}$ represents

(1) T^2

(2) T^{-2}

(3) $T^{1/2}$

(4) $T^{3/2}$

3. Given below are two statements:
Statement I: The force of attraction between a solid spherical body of uniform mass density and a point mass m situated inside is just as if the entire mass of the solid body is concentrated at its centre.
Statement II: Force on a point mass m situated inside a spherical shell is zero.
In the light of the above statements, choose the most appropriate answer from the options given below.

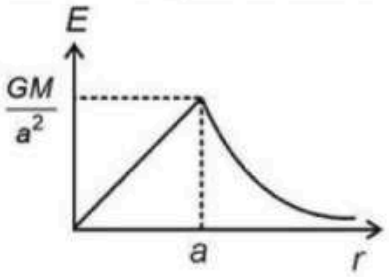
(1) Both statement I and statement II are correct

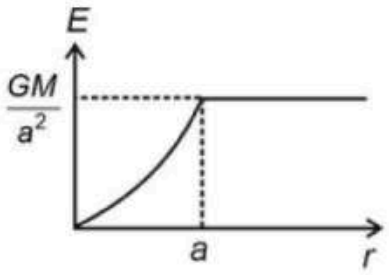
(2) Both statement I and statement II are incorrect

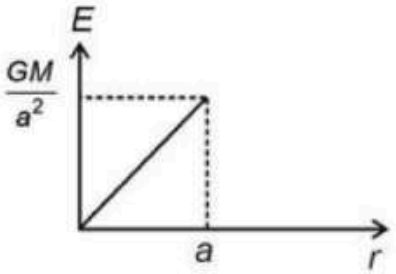
(3) Statement I is correct while statement II is incorrect

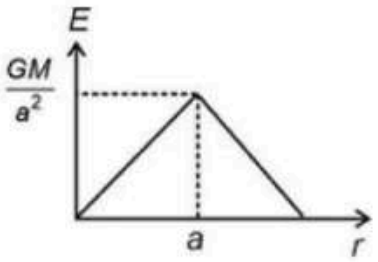
(4) Statement I is incorrect while statement II is correct

4. Choose the correct variation of gravitational field intensity (E) due to a uniform solid sphere with the distance from the centre (r) of sphere (where a is the radius of sphere)

(1)


(2)


(3)


(4)


5. If a projectile is launched from a point at a height "h" from surface of earth in a direction tangent to earth surface with speed of $\sqrt{\frac{3GM}{2(R+h)}}$, what will be path followed by it?

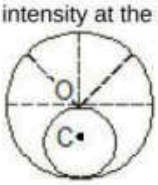
(1) Circle

(2) Parabola

(3) Ellipse

(4) Hyperbola

6. A cavity of radius $\frac{R}{2}$ is made inside a solid sphere of mass M and radius R (as shown in figure). The gravitational field intensity at the center of cavity is



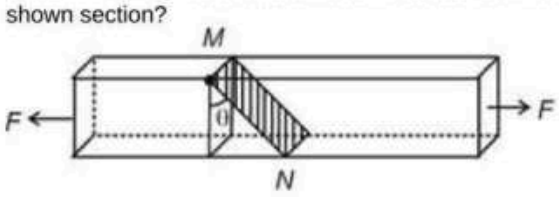
(1) Zero

(2) $\frac{GM}{2R^2}$

(3) $\frac{2GM}{R^2}$

(4) $\frac{GM}{R^2}$

7. A rod, shown in figure, is subjected to equal and opposite forces. MN is a plane making an angle $\theta = 45°$ with cross-section A of the rod. What is tensile stress on MN shaded shown section?



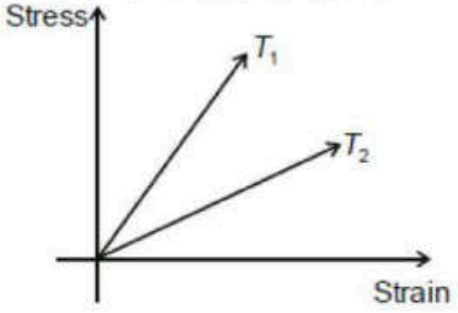
(1) $\frac{\sqrt{2}F}{A}$

(2) $\frac{F}{2A}$

(3) $\frac{F}{A}$

(4) $\frac{2F}{A}$

8. The stress-strain graph for a metallic wire is shown at two different temperature T_1 and T_2. Then



(1) $T_1 > T_2$

(2) $T_2 > T_1$

(3) $T_1 = T_2$

(4) Can't predict

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9. The Poisson's ratio of the material of cylindrical wire when density of the material of wire is constant is

(1) – 0.5

(2) + 0.5

(3) Zero

(4) –1

10. A diver is 20 m below the surface of water. The pressure experienced by the diver is about

(1) 2×10^5 Pa

(2) 1×10^5 Pa

(3) 3×10^5 Pa

(4) Zero

11. Consider a horizontally oriented syringe containing water. The diameter of plunger is 4 mm and the diameter of nozzle is 2 mm. The plunger is pushed with a constant speed of 0.5 m s^{-1}. The speed of water coming out of nozzle is

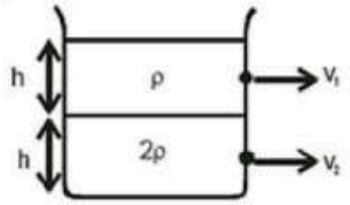
(1) 2 m s^{-1}

(2) 4 m s^{-1}

(3) 8 m s^{-1}

(4) Zero

12. Equal volume of two immiscible liquids of densities ρ and 2ρ are filled in a vessel as shown. Two small holes are punched at depth $\frac{h}{2}$ and $\frac{3h}{2}$ from the surface of lighter liquid, if v_1 and v_2 are the velocities of efflux at these two holes then $\frac{v_1}{v_2}$ is



(1) $\frac{1}{2}$

(2) $\frac{1}{\sqrt{2}}$

(3) $\frac{1}{2\sqrt{2}}$

(4) $\frac{1}{4}$

13. Two identical metal balls when fall in a liquid move with constant speed *v*. If both the balls are melted and combined to form a single ball and allowed to fall through the same liquid then the constant speed with which the ball will move is

(1) $2^{\frac{4}{3}}v$

(2) $2^{\frac{2}{3}}v$

(3) $2^{\frac{1}{3}}v$

(4) $2^{\frac{5}{3}}v$

14. If a section of a thin soap bubble (of radius *R*) through its centre is considered, then force on one half of bubble due to surface tension is

(1) $\frac{2T}{R}$

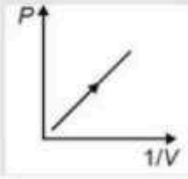
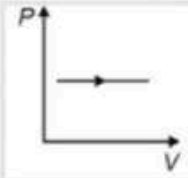
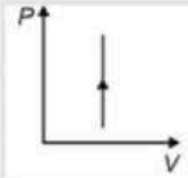
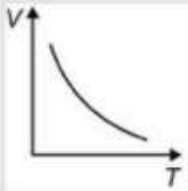
(2) $4\pi RT$

(3) $\frac{T}{R}$

(4) $\frac{4T}{R}$

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15. Column-I comprises of different graphs for an ideal gas and column-II contains statements. Match the column-I with column-II [*P* is pressure and *V* is volume]

	Column-I		Column-II
a.		(p)	Volume is constant
b.		(q)	Temperature is constant
c.		(r)	Pressure is constant
d.		(s)	Adiabatic process

(1) a(q); b(p); c(r); d(s)

(2) a(q); b(r); c(p); d(s)

(3) a(p); b(q); c(s); d(r)

(4) a(r); b(p); c(s); d(q)

16. **Assertion (A):** If same amount of heat is given to solid sphere and hollow spheres of same material having same radius at a particular temperature, the rise in temperature of the hollow sphere will be more.
Reason (R): Hollow sphere has less mass as compared to solid one of same material and same size and hence its heat capacity is lesser.
In the light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true and (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) A is false but (R) is true

17. 30 g ice at 0°C is mixed with 2 g steam at 100°C. The final temperature of the mixture is

(1) 0°C

(2) 100°C

(3) 50°C

(4) 65°C

18. A black body has maximum intensity wavelength λ_m at 1000 K. Then wavelength of maximum intensity at 3000 K is

(1) $3\lambda_m$

(2) λ_m

(3) $\frac{2}{3}\lambda_m$

(4) $\frac{\lambda_m}{3}$

19. The ratio of radiant energies radiated per unit surface area by two black bodies is 16 : 1. The temperature of hotter body is 1000 K, then the temperature of the colder body will be

(1) 250 K

(2) 500 K

(3) 1000 K

(4) 62.5 K

20. If a liquid takes 20 s in cooling from 80°C to 70°C and 50 s in cooling from 60°C to 50°C, then the surrounding temperature will be (assuming Newton law of cooling is valid)

(1) 41.7°C

(2) 45°C

(3) 30°C

(4) 25°C

21. The molar heat capacity in a process of a diatomic gas if it does a work of $\frac{Q}{4}$ when a heat of *Q* is supplied to it is

(1) $\frac{5}{2}R$

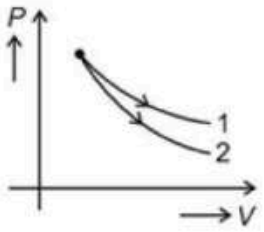
(2) $\frac{10}{3}R$

(3) $\frac{6}{7}R$

(4) $\frac{2}{5}R$

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22. Two ideal gases undergo adiabatic expansion, for which the corresponding *P-V* diagram is drawn as shown in the figure. The plot 1 and 2 should respectively correspond to



(1) N_2, He

(2) O_2, N_2

(3) He, O_2

(4) N_2, CO

23. Heat is supplied to a diatomic gas at constant pressure. The ratio of $\Delta Q : \Delta U : \Delta W$ is

(1) 5 : 3 : 2

(2) 5 : 2 : 3

(3) 7 : 5 : 2

(4) 7 : 2 : 5

24. The efficiency of carnot engine operating between reservoirs maintained at 27°C and 327°C is

(1) 0.75

(2) 0.60

(3) 0.80

(4) 0.50

25. The respective ratio of rms speed of O_2 and Ar gas molecules kept at same volume with temperature ratio 1 : 5 is (take molar mass of Ar as 40 g)

(1) 2 : 1

(2) 1 : 2

(3) 4 : 1

(4) 1 : 4

26. If degrees of freedom of a gas are *n*, then the value of $\left(\frac{C_P}{C_V} - 1\right)$ is

(1) *n*

(2) $\frac{n}{2}$

(3) $\frac{1}{n}$

(4) $\frac{2}{n}$

27. A non linear triatomic gas is filled inside a vessel. If 40% of moles dissociates into individual atoms then degree of freedom for gas mixture is

(1) 3.75

(2) 4

(3) 4.25

(4) 3

28. A particle is executing SHM along x-axis about origin with amplitude *A* and time period *T*. The time taken by the particle in going from $x = 0$ to $x = \frac{A}{2}$ will be

(1) $\frac{T}{6}$

(2) $\frac{T}{12}$

(3) $\frac{T}{14}$

(4) $\frac{T}{21}$

29. A particle of mass 5 g is undergoing SHM of amplitude 6 cm and time period 0.5 s. The maximum value of restoring force acting on the particle is [$\pi^2 = 10$]

(1) 10^{-10} N

(2) 4.8×10^{-2} N

(3) 60 N

(4) 4.08×10^{-3} N

30. A body executes S.H.M. with an amplitude *A*. At what displacement from the mean position, potential energy of the body is $\frac{3}{4}$th of its total energy?

(1) $\frac{A}{2}$

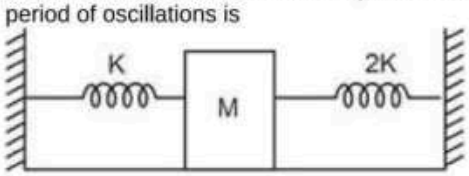
(2) $\frac{A}{4}$

(3) $\frac{\sqrt{3}}{2}A$

(4) $\frac{A}{\sqrt{2}}$

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31. Two identical springs of spring constant K and 2K are attached to a block of mass *M* and to fixed supports as given in figure. If the mass is displaced from its equilibrium position on either side, it executes a simple harmonic motion. The period of oscillations is



(1) $T = 2\pi\sqrt{\frac{M}{2K}}$

(2) $T = 2\pi\sqrt{\frac{M}{K}}$

(3) $T = 2\pi\sqrt{\frac{3M}{2K}}$

(4) $T = 2\pi\sqrt{\frac{M}{3K}}$

32. A ring of radius 20 cm is hinged from a point on its circumference and oscillates in vertical plane. The time period of oscillation is

(1) 2 s

(2) $\frac{\pi}{5}$ s

(3) $\frac{2\pi}{5}s$

(4) $\frac{\sqrt{2}\pi}{5}$ s

33. If $\vec{u}$ is instantaneous velocity of a medium particle of transverse wave and $\vec{v}$ is velocity of transverse wave, then

(1) $\vec{u} \cdot \vec{v} = 0$

(2) $\left|\vec{u} \times \vec{v}\right| = 0$

(3) $\left|\vec{u}\right| = \left|\vec{v}\right|$ always

(4) $\left|\vec{u}\right| > \left|\vec{v}\right|$ always

34. A wave travelling in the positive *y*-direction has displacement along *x*-direction. If its amplitude is 1 m, wavelength is π m and frequency is $\frac{1}{\pi}$ Hz then wave can be represented by

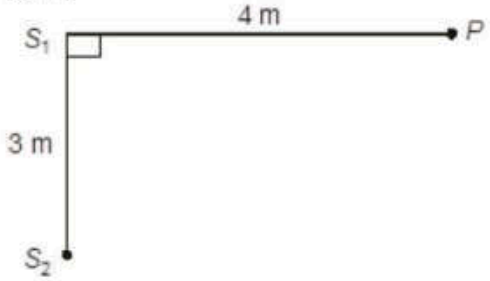
(1) $X = \sin(2\pi t - 2\pi y)$

(2) $X = \sin(2t - 2y)$

(3) $Y = \sin(t - x)$

(4) $Y = \sin(2\pi t - 2\pi x)$

35. S_1 and S_2 are two coherent sources of sound of frequency 110 Hz each. They have no initial phase difference. The intensity at a point *P* due to S_1 is I_0 and due to S_2 is $4I_0$. If the velocity of sound is 330 m/s, then the resultant intensity at *P* is



(1) I_0

(2) $9I_0$

(3) $3I_0$

(4) $8I_0$

36. Fundamental frequency of open organ pipe of length *L* is *f*. If 20% of its length is dipped in water, then fundamental frequency of remaining length is

(1) $\frac{5}{8}f$

(2) $\frac{4f}{5}$

(3) $\frac{3}{4}f$

(4) $\frac{7}{5}$

37. A string of length 1 m and mass 5 g is fixed at both ends. The tension in the string is 8.0 N. The string is set into vibration using an external vibrator of frequency 80 Hz. The separation between successive nodes on the string is

(1) 16.6 cm

(2) 25 cm

(3) 50 cm

(4) 33.3 cm

38. Two tuning forks *A* and *B* of frequencies 300 Hz and '*f*' Hz respectively are sounded together to produce beat frequency of 5 Hz. If fork *A* is filed then beat frequency gets reduced. Then *f* (in Hz) is

(1) 300

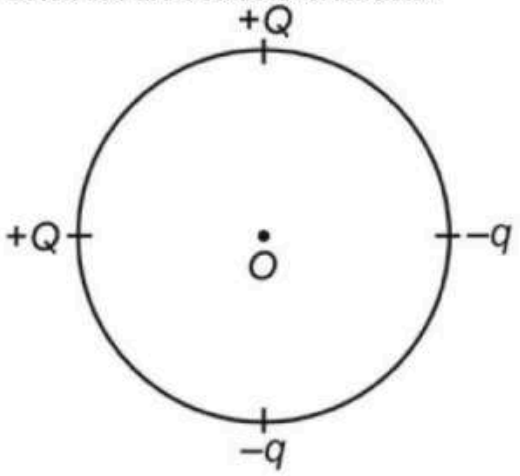
(2) 295

(3) 305

(4) 310

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39. Four charges are placed at the circumference of a dial clock as shown in the figure. What is the time shown by the clock if the hour hand of the clock points in the direction of net electric field at the centre '*O*' of the clock?



(1) 6 : 00

(2) 7 : 30

(3) 4 : 30

(4) 10 : 30

40. Choose the correct statement among the following.

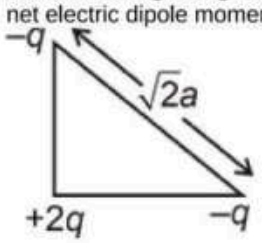
(1) Inside a conductor, electrostatic field is zero

(2) At the surface of charged conductor, electrostatic field must be normal to the surface every point

(3) The interior of a conductor can have no excess charge in the static condition

(4) All of these

41. Three point charges $+2q$, $-q$ and $-q$ are placed at the corners of a right angled triangle as shown in the figure. The net electric dipole moment of the system will be



(1) $\sqrt{2}\,qa$

(2) $2qa$

(3) qa

(4) Zero

42. Potential and field strength at a certain distance from a point charge are 250 V and 50 N/C. Distance of the point from the charge is

(1) 1 m

(2) 2 m

(3) 3 m

(4) 5 m

43. Magnitude of electric field inside a charged conducting shell having uniform surface charge density is

(1) Proportional to surface charge density

(2) Proportional to charge

(3) Zero

(4) Non-zero and constant

44. When a soap bubble is given a positive charge then its size increases. If instead of positive, the soap bubble is given a negative charge then its size

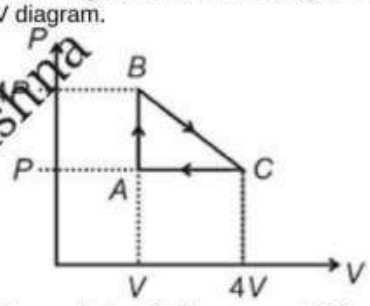
(1) Decreases

(2) Increases

(3) May increase or decrease

(4) Remains unchanged

45. An ideal gas is taken around cycle ABCA as shown in the P-V diagram.



The work done in the process BC is

(1) $\frac{13}{2}PV$

(2) $\frac{15}{2}PV$

(3) $\frac{11}{2}PV$

(4) $7PV$

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CHEMISTRY

46. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**:
Assertion (A) : Decrease in free energy causes spontaneous reaction.
Reason (R) : Spontaneous reactions are invariably endothermic reaction.
In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

47. Mole of oxalate ions required to reduce one mole of permanganate ion in acidic medium is

(1) $\frac{2}{3}$

(2) $\frac{3}{2}$

(3) $\frac{5}{2}$

(4) $\frac{2}{5}$

48. For an ideal solution, the correct option is

(1) $\Delta_{mix}S = 0, \Delta_{mix}V = 0, \Delta_{mix}H = 0, \Delta_{mix}G > 0$

(2) $\Delta_{mix}S > 0, \Delta_{mix}V = 0, \Delta_{mix}H = 0, \Delta_{mix}G < 0$

(3) $\Delta_{mix}S < 0, \Delta_{mix}V = 0, \Delta_{mix}H = 0, \Delta_{mix}G = 0$

(4) $\Delta_{mix}S > 0, \Delta_{mix}V = 0, \Delta_{mix}H < 0, \Delta_{mix}G < 0$

49. Consider the following two statements.
Statement (I): Galvanic cells that are designed to convert the energy of combustion of fuels like hydrogen, methane, etc directly into electrical energy are called fuel cells.
Statement (II): A secondary cell after use can be recharged by passing current through it in the opposite direction.
In the light of above two statements, choose the correct option.

(1) Both statement (I) and statement (II) are correct

(2) Statement (I) is correct and statement (II) is incorrect

(3) Statement (I) is incorrect and statement (II) is correct

(4) Both statement (I) and statement (II) are incorrect

50. Match List-I with List-II for given balanced chemical reaction
$a\,MnO_4^- + b\,C_2O_4^{2-} + c\,H^+ \rightarrow a\,Mn^{2+} + d\,CO_2 + e\,H_2O$

List-I	List-II
(a) a	(i) 5
(b) b	(ii) 2
(c) c	(iii) 16
(d) d	(iv) 10

Correct match is

(1) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

(2) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

(3) (a)-(i), (b)-(ii), (c)-(iv), (d)-(iii)

(4) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

51. Consider the following statements
(a) The solutions which show large negative deviation from Raoult's law form minimum boiling azeotrope.
(b) Two solutions having same osmotic pressure at a given temperature are called isotonic solution.
(c) K_f is independent of solute concentration.
The correct statement(s) are

(1) (a) and (b) only

(2) (b) and (c) only

(3) (a) and (c) only

(4) (a), (b) and (c)

52. Which of the following mixture of solutions can behave as a buffer solution?

(1) 10 mL of 1 M HCl + 10 mL of 1 M NH_4OH

(2) 10 mL of 1 M HCl + 10 mL of 0.5 M NH_4OH

(3) 10 mL of 0.5 M HCl + 10 mL of 0.5 M NH_4OH

(4) 10 mL of 0.5 M HCl + 10 mL of 1 M NH_4OH

53. Match list-I with list-II and choose the correct option.

	List-I (Reactions)	**List-II (Relation between ΔH and ΔU)**
a.	$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$	(i) $\Delta H = \Delta U - \frac{1}{2}RT$
b.	$PCl_5(g) \rightarrow PCl_3(g) + Cl_2(g)$	(ii) $\Delta H = \Delta U + RT$
c.	$C(s) + O_2(g) \rightarrow CO_2(g)$	(iii) $\Delta H = \Delta U - 2RT$
d.	$CO(g) + \frac{1}{2}O_2(g) \rightarrow CO_2(g)$	(iv) $\Delta H = \Delta U$

(1) a(iii), b(iv), c(i), d(ii)

(2) a(ii), b(iii), c(iv), d(i)

(3) a(ii), b(iv), c(i), d(iii)

(4) a(iii), b(ii), c(iv), d(i)

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54. Consider the following equilibrium,
$NH_4COONH_2(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$
if the equilibrium pressure for the above reaction is 2.1 atm, then K_p for the reaction will be

(1) 0.686 atm^3

(2) 1.372 atm^3

(3) 2.744 atm^3

(4) 5.488 atm^3

55. Given below are two statements
Statement I: Magnitude of work done in adiabatic expansion is more than work done in isothermal expansion.
Statement II: If temperature of a crystalline solid is raised from 0 K to 115 K, entropy increases.
In the light of above statements, choose the most appropriate answer from the options given below

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are correct

(4) Both statement I and statement II are incorrect

56. pH of 0.04 M Calcium acetate solution is (pk_a of CH_3COOH = 4.74)

(1) 4.42

(2) 10.25

(3) 8.82

(4) 7.25

57. If bond dissociation enthalpies of H_2, Br_2 and HBr respectively are 435 kJ mol^{-1}, 192 kJ mol^{-1} and 368 kJ mol^{-1}, then the enthalpy change during the reaction $H_2(g) + Br_2(g) \rightarrow 2HBr(g)$, is

(1) –137 kJ

(2) –109 kJ

(3) –259 kJ

(4) 259 kJ

58. Match species in List-I with their conjugate acid/base in List-II.

List-I		List-II
a. $H_2PO_4^-$	(i)	$H_2PO_2^-$
b. HPO_4^{2-}	(ii)	$H_2PO_3^-$
c. HPO_3^{2-}	(iii)	PO_4^{3-}
d. H_3PO_2	(iv)	H_3PO_4

Correct match is

(1) a(iv), b(iii), c(ii), d(i)

(2) a(iv), b(i), c(iii), d(ii)

(3) a(iv), b(iii), c(i), d(ii)

(4) a(i), b(iv), c(iii), d(ii)

59. Which among the following is best reducing agent in aqueous medium?

(1) Na

(2) Li

(3) Cu

(4) F_2

60. Given below are two statements
Statement I: Elevation of boiling point (ΔT_b) depends on the nature of solute particles.
Statement II: Ebullioscopic constant (K_b) depends on the nature of the solvent.
In the light of above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are true

(2) Both statement I and statement II are false

(3) Statement I is true but statement II is false

(4) Statement I is false but statement II is true

61. Which of the following expression correctly represents the equivalent conductance of $Al_2(SO_4)_3$ at infinite dilution? (Given that $\lambda^o_{Al^{3+}}$ and $\lambda^o_{SO_4^{2-}}$ are the molar conductance at infinite dilution of the respective ions)

(1) $3\lambda^o_{Al^{3+}} + 2\lambda^o_{SO_4^{2-}}$

(2) $\frac{1}{6}\lambda^o_{Al^{3+}} + \frac{1}{6}\lambda^o_{SO_4^{2-}}$

(3) $\frac{1}{3}\lambda^o_{Al^{3+}} + \frac{1}{2}\lambda^o_{SO_4^{2-}}$

(4) $\lambda^o_{Al^{3+}} + \lambda^o_{SO_4^{2-}}$

62. Solubility of $Ni(OH)_2$ in a solution of pH = 11 will be {K_{sp} of $Ni(OH)_2 = 2 \times 10^{-15}$}

(1) 2×10^{-9} M

(2) 2×10^{-12} M

(3) 4.4×10^{-8} M

(4) 4.4×10^{-10} M

63. Given below are two statements one is labelled as Assertion (A) and other is labelled as Reason (R).
Assertion (A): ClO_4^- cannot undergo disproportionation reaction.
Reason (R): Cl is at it's highest oxidation state in ClO_4^- ion.
In the light of above statements, choose the most appropriate answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

Vivek Krishna

64. 0.2 molal aqueous solution of $K_3[Fe(CN)_6]$ ionizes to 40%. Depression in freezing point of solution will be ($K_f(H_2O)$ = 1.86 K molal^{-1})

(1) 0.82 K

(2) 0.45 K

(3) 1.23 K

(4) 0.63 K

65. Given below are two statements:
Statement I: When a system is at equilibrium, the entropy is maximum.
Statement II: When a system is at equilibrium, ΔS_{total} is zero
In the light of the above statements, choose the *most appropriate* answer from the options given below.

(1) Both Statement I and Statement II are incorrect

(2) Both Statement I and Statement II are correct

(3) Statement I is correct but Statement II is incorrect

(4) Statement I is incorrect but Statement II is correct

66. Oxidation state of P in PO_4^{3-}, of S in SO_4^{2-} and that of Cr in $Cr_2O_7^{2-}$ respectively are

(1) +3, +6 and +5

(2) +5, +3 and +6

(3) +3, +6 and +6

(4) +5, +6 and +6

67. Total number of Faraday's required to produce 1.08 g of Ag from aq AgCl is (Atomic Mass of Ag = 108 u)

(1) 0.02 F

(2) 1 F

(3) 0.1 F

(4) 0.01 F

68. If the mass percentage of urea in an aqueous solution is 20% then molality of the solution will be

(1) 5.25 m

(2) 2.35 m

(3) 4.17 m

(4) 4.31 m

69. When 6 moles of He gas at 27°C undergoes reversible isothermal expansion from 30 L to 60 L, then work done will be

(1) –4.5 kJ

(2) 4.5 kJ

(3) –10.4 kJ

(4) 10.3 kJ

70. Disproportionation reaction among the following is,

(1) $KClO_3 \xrightarrow{\Delta} KCl + \frac{3}{2}O_2$

(2) $2H_2 + O_2 \rightarrow 2H_2O$

(3) $Cl_2 + 2NaOH \rightarrow NaOCl + NaCl + H_2O$

(4) $NaOH + HNO_3 \rightarrow NaNO_3 + H_2O$

71. Which of the following is not an intensive property?

(1) Mass per unit volume

(2) Volume per unit mass

(3) Gibbs energy

(4) Pressure

72. The standard Gibbs free energy change for the given cell reaction (in kJ mol^{-1}) will be..
$Zn(s) + 2Ag^+(aq) \rightarrow Zn^{2+}(aq) + 2Ag(s)$
$(E^\circ_{Zn^{2+}/Zn} = -0.76\,V;\ E^\circ_{Ag^+/Ag} = 0.80\,V)$

(1) – 25.35

(2) – 150.21

(3) – 301.08

(4) – 501.12

73. The oxidation number of O in $O_2F_2$is

(1) +2

(2) –2

(3) –1

(4) +1

74. The pH of the solution obtained by adding 50 mL, 0.1 M NaOH to 60 mL, 0.2 M CH_3COOH solution is (pK_a of CH_3COOH = 4.75, log 5 = 0.699, log 7 = 0.845)

(1) 4.75

(2) 4.60

(3) 3.50

(4) 5.25

75. The EMF of the cell, Fe I Fe^{+2} (1M) || Ag^+ (1M) | Ag is +1.24 V at 25°C. If $E^\circ_{Ag^+/Ag}$ = 0.80 V, then $E^\circ_{Fe^{2+}/Fe}$ will be

(1) 0.44 V

(2) – 2.04 V

(3) 2.04 V

(4) – 0.44 V

Vivek Krishna

76. If two volatile liquids A(P_A° = 540 torr) and B(P_B° = 420 torr) are mixed in the molar ratio of 2 : 1 then total vapour pressure of solution is

(1) 100 torr

(2) 200 torr

(3) 600 torr

(4) 500 torr

77. Which of the following forms azeotropic mixture?

(1) 90% ethanol + 10% water

(2) 75% nitric acid + 25% water

(3) 80% ethanol + 20% water

(4) 68% nitric acid + 32% water

78. The oxidation number of central Br atom in Br_3O_8 is

(1) 0

(2) +2

(3) +4

(4) +6

79. Given below are two statements, one is assertion (A) other is reason (R).
Assertion (A) : The freezing point of 0.1 m glucose solution is higher than that of 0.1 m NaCl solution.
Reason (R) : Depression in freezing point is inversly proportional to the number of moles of non-volatile solute particles present in the solution.
In the light of above statements, choose the correct answer.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

80. Consider the following equilibrium reactions
(a) $CH_3COOC_2H_5(aq) + H_2O(l) \rightleftharpoons CH_3COOH(aq) + C_2H_5OH(aq)$

(b) $Fe^{3+}(aq) + SCN^-(aq) \rightleftharpoons [Fe(SCN)]^{2+}(aq)$
(c) $H_2O(l) \rightleftharpoons H_2O(g)$

(d) $Ca(OH)_2(s) + (aq) \rightleftharpoons Ca^{2+}(aq) + 2OH^-(aq)$
Which among the given reactions represent heterogeneous equilibria?

(1) (c) and (d) only

(2) (b), (c) and (d) only

(3) (a) and (b) only

(4) (a), (c) and (d) only

81. Heat of neutralisation of a weak dibasic acid is found to be – 25.2 kcal mol^{-1}. The enthalpy of ionisation of weak acid will be

(1) 2.2 kcal mol^{-1}

(2) 13.3 kcal mol^{-1}

(3) 6.2 kcal mol^{-1}

(4) 1.5 kcal mol^{-1}

82. The mass of urea to be dissolved in 100 g of water (K_b = 0.52 K kg mol^{-1}) to prepare a solution having boiling point 101°C, is

(1) 10.26 g

(2) 4.2 g

(3) 11.54 g

(4) 14.66 g

83. Given below are two statements.
Statement (I): $Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$ is a displacement reaction.
Statement (II): Average oxidation number of bromine in tribromooctaoxide is $\frac{16}{3}$.
In the light of above statements choose the **correct** option.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

84. The electrode potential of hydrogen electrode is when H_3O^+ ion concentration is 10^{-5} M (pH_2 = 1 atm)

(1) – 0.059 V

(2) – 0.295 V

(3) – 0.118 V

(4) – 0.59 V

85. On dilution of strong electrolyte

(1) Λ_m increases while κ decreases

(2) Λ_m decreases while κ increases

(3) Both Λ_m and κ decreases

(4) Both Λ_m and κ increases

86. For which of the following substance, standard enthalpy of formation is zero?

(1) C(diamond)

(2) $Br_2(g)$

(3) C(graphite)

(4) $I_2(l)$

Vivek Krishna

87. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A) : Addition of Ar to equilibrium $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ at constant volume will shift equilibrium to backward direction.
Reason (R) : Ar only reacts with NH_3
In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both (A) and (R) are true and the (R) is the correct explanation of the (A)

(2) Both (A) and (R) are true but the (R) is not the correct explanation of the (A)

(3) (A) is true statement but (R) is false

(4) Both (A) and (R) are false statements

88. Match reaction in List-I with their correct relation of K_p and K_c in List-II

List-I	List-II
(a) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$	(i) $K_p = K_c(RT)^{-2}$
(b) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$	(ii) $K_p = K_c(RT)$
(c) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$	(iii) $K_p = K_c$
(d) $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$	(iv) $K_p = K_c(RT)^{-1}$

Choose the **correct** match

(1) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

(2) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)

(3) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)

(4) (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv)

89. Match batteries in list-1 with their electrolyte in List-2.

	List-1		List-2
(a)	Dry cell	(i)	38% aq H_2SO_4
(b)	Mercury cell	(ii)	Paste of NH_4Cl and $ZnCl_2$
(c)	(H_2–O_2) Fuel cell	(iii)	Paste of KOH and ZnO
(d)	Lead storage battery	(iv)	Conc. (aq) NaOH

(1) (a) (ii), (b) (iii), (c)(i), (d)(iv)

(2) (a) (iii), (b) (ii), (c)(i), (d) (iv)

(3) (a)(iii), (b) (ii), (c) (iv), (d) (i)

(4) (a) (ii), (b) (iii), (c) (iv), (d) (i)

90. Given below are two statements
Statement I: In corrosion, at anode, Fe gets oxidised to Fe^{2+} ion.
Statement II: In corrosion, at cathode, $O_2(g)$ reduced in presence of H^+ ions into H_2O.
In the light of above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are true

(2) Statement I is true but statement II is false

(3) Statement I is false but statement II is true

(4) Both statement I and statement II are false

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91. To minimise water loss, grass leaves curl inward because

(1) Of bulliform cells, present on abaxial surface

(2) Of trichomes present on abaxial surface

(3) They have less number of stomata on upper surface

(4) Of large, empty colourless cells on adaxial surface

92. The epidermal hairs on stem are

(1) Usually unicellular

(2) Not secretory in function

(3) Helpful in preventing water loss due to transpiration

(4) Always unbranched and soft

93. Read the given statements and choose the **correct** option.
Statement A : The gametophyte of pteridophytes requires warm and humid places to grow.
Statement B : Majority of pteridophytes are homosporous.

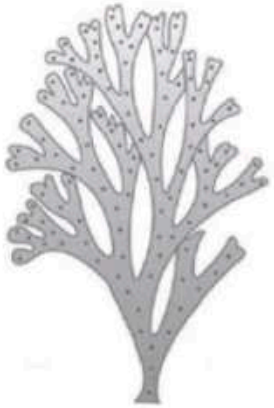
(1) Only statement A is correct

(2) Only statement B is correct

(3) Both statements A and B are correct

(4) Both statements A and B are incorrect

94.



Identify the organism in the given figure and select the correct feature of the class to which this organism belongs.

(1) Mostly found in fresh water

(2) Pear shaped gametes

(3) Pyrenoids stores food in them

(4) Sexual reproduction only by oogamy

95. Stamens are united into more than two bundles. This type of cohesion of stamens is known as

(1) Monoadelphous

(2) Diadelphous

(3) Polyandrous

(4) Polyadelphous

96. Classification for flowering plants given by George Bentham and Joseph Dalton Hooker

(1) Is also known as artificial system of classification

(2) Based mainly on the androecium structure

(3) Considers embryology, ultrastructure, phytochemistry

(4) Uses only gross superficial morphological characters

97. The anatomical features which are common to both monocot stem and dicot leaf are the presence of

(1) Endodermis and sub-stomatal cavity

(2) Conjoint and closed vascular bundle

(3) Palisade and spongy mesophyll parenchyma

(4) Phloem parenchyma and water containing cavities in vascular bundles

98. The earliest system of classification of organisms

(1) Used only gross superficial morphological characters

(2) Were based on natural affinities among the organisms

(3) Considered both external and internal features

(4) Were based on evolutionary relationships and cytological informaton.

99. In gymnosperms the male gametophyte is highly reduced, known as

(1) Endosperm

(2) Embryo sac

(3) Nucellus

(4) Pollen grain

100. Coconut fruit differs from mango as the former,

(1) Is a drupe and latter is a berry

(2) Has edible mesocarp

(3) Has hard and stony endocarp

(4) Has edible endosperm

101. In dicotyledonous plants, lateral roots arise from

(1) Cortex

(2) Medullary rays

(3) Endodermis

(4) Pericycle

102. Select the **incorrectly** matched pair.

(1) Pulvinus – Swollen leaf base in some leguminous plants

(2) Bracts – Reduced leaf found at the base of pedicel

(3) Receptacle – Swollen end of the stalk bearing successively arranged floral whorls

(4) Asymmetric flower – Found in gulmohur and *Cassia*

(1) (1)

(2) (2)

(3) (3)

(4) (4)

103. Select the **odd** one out w.r.t. alternate phyllotaxy.

(1) China rose

(2) Mustard

(3) Sunflower

(4) Guava

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104. Read the given statements of assertion (A) and reason (R) and choose the **correct** option.
Assertion (A): Stems of maize and wheat do not possess the ability to form secondary xylem and secondary phloem tissues.
Reason (R): In stems of maize and wheat, the xylem and phloem within a vascular bundle are arranged in an alternate manner along the different radii.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) Both (A) and (R) are false.

(4) (A) is true but (R) is false.

105. Choose the **correct** option after reading the following statements.
Statement A: In dicot seeds an embryo has two cotyledons along with plumule and radicle.
Statement B: In seeds of cereals, the seed coat is membranous and generally fused with the fruit wall.

(1) Only statement A is incorrect

(2) Only statement B is incorrect

(3) Both the statements are incorrect

(4) Both the statements are correct

106. The only living component of xylem is

(1) Tracheid

(2) Vessel

(3) Xylem parenchyma

(4) Xylem fibres

107. Select the **odd** one out w.r.t ground tissue system.

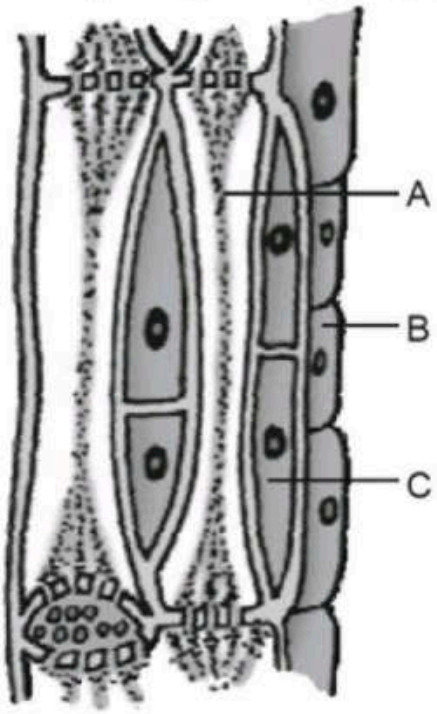
(1) Pith

(2) Medullary rays

(3) Vascular bundles

(4) Pericycle

108. In below given diagram correctly identify A, B and C.



(1) A $\longrightarrow$ Sieve tube
C $\longrightarrow$ Phloem parenchyma

(2) A $\longrightarrow$ Sieve tube element
B $\longrightarrow$ Phloem parenchyma

(3) A $\longrightarrow$ Companion cell
B $\longrightarrow$ Albuminous cell

(4) A $\longrightarrow$ Phloem fibre
B $\longrightarrow$ Companion cell

109. Sexual reproduction in red algae is different from brown algae because in the former

(1) It is isogamous with complex pre fertilisation development

(2) It is anisogamous with simple post fertilisation development

(3) It is oogamous with complex post fertilisation development

(4) It may be isogamous, anisogamous or oogamous

110. All gymnosperms

(1) Are heterosporous

(2) Produce male and female cones on separate trees

(3) Have needle-like leaves

(4) Have coralloid roots

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111. In which among the following flowers, adhesion of stamens are **not** observed?

(1) Pea

(2) Potato

(3) Lily

(4) Brinjal

112. In gymnosperms, pollen grains are carried to the opening of ovules by

(1) Insects

(2) Air current

(3) Water current

(4) Higher animals

113. Members of Rhodophyceae differs from Phaeophyceae, as the former

(1) Has two, unequal laterally attached flagella

(2) Are primarily found in marine habitats

(3) Contains cell wall composed of cellulose, pectin and polysulphate esters

(4) Involves non motile spores for sexual reproduction

114. Read the following statements and mark true **(T)** or false **(F)** then choose the **correct** option.
A. Gymnosperms generally have tap root.
B. Prothallus is usually dependent on sporophyte body.
C. In mosses, archegonia are produced at the apex of the leafy shoots.

(1) **A**-T, **B**-F, **C**-T

(2) **A**-T, **B**-T, **C**-T

(3) **A**-F, **B**-T, **C**-T

(4) **A**-T, **B**-F, **C**-F

115. Syncarpous gynoecium is seen in

(1) Mustard and tomato

(2) Mustard and lotus

(3) Lotus and rose

(4) Rose and tomato

116. Mark the **incorrect** option w.r.t. the symbols used in floral formula of the members of family Brassicaceae.

(1) $\oplus$ ♂

(2) K_{2+2}

(3) $C_{\times 4}\ A_{2+4}$

(4) $\overline{G}_2$

117. In *Laminaria*, the function of frond is similar to that of

(1) Root in angiosperms

(2) Stem of gymnosperms

(3) Branch in angiosperm

(4) Leaf in angiosperms

118. Which of the following statements is **correct** w.r.t isobilateral leaf?

(1) The adaxially placed palisade parenchyma is made up of elongated cells.

(2) The stomata are present on both the surfaces of the epidermis; and the mesophyll is not differentiated into palisade and spongy parenchyma.

(3) Reticulate type of venation is present.

(4) The oval or round and loosely arranged spongy parenchyma cells are situated below the palisade cells.

119. Select the **incorrect** statement w.r.t. epiblema

(1) It is usually single layered

(2) Their outer walls protrude outwards in the form of root hairs.

(3) It is the outermost layer of dicot root.

(4) It possesses casparian strips

120. Kelps are

(1) Filamentous red algae

(2) Profusely branched brown algae

(3) Profusely branched red algae

(4) Colonial green algae

121. Read the following statements (A-D).
(A) In *Dianthus* and *Primrose*, ovules are borne on central axis and septa are absent.
(B) Variation in the length of filaments of stamens in flower is seen in *Salvia* and mustard.
(C) In sunflower and marigold, the placenta forms a ridge along the ventral suture of the ovary and ovules are borne on this ridge.
(D) In mustard, ovary is one-chambered but it becomes two chambered due to the formation of false septum.
The **correct** statements are

(1) Only (A) and (D)

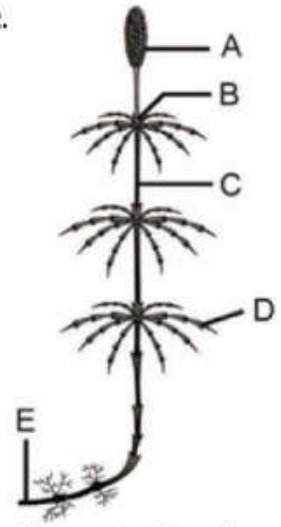
(2) Only (B), (C) and (D)

(3) Only (A), (B) and (D)

(4) (A) and (C)

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122.



Observe the above diagram and find the **correct** statement.

(1) E is a type of root

(2) B is a part of leaf

(3) A is composed of loosely arranged sporophylls

(4) Ploidy of cells of C is diploid

123. Lax is

(1) Compact strobilus or cone in bryophytes

(2) Sporangium having spores

(3) Spirally arranged sporophylls along an axis

(4) Female gametophyte in gymnosperms

124. Which of the given algae is used as food supplements even by space travellers?

(1) *Chara*

(2) *Chlorella*

(3) *Ulothrix*

(4) *Polysiphonia*

125. In homosporous species of pteridophytes, gametophyte (prothallus) is

(1) Monoecious

(2) Dioecious

(3) Found on parent sporophyte

(4) Single celled

126. ____(X)____ uses the chemical constituents of plants to resolve confusions in classification.
Select the correct option for (X).

(1) Cytotaxonomy

(2) Karyotaxonomy

(3) Numerical taxonomy

(4) Chemotaxonomy

127. Match the classes of pteridophyta given in **column I** with their respective members given in **column II**

	Column I		**Column II**
a.	Psilopsida	(i)	*Selaginella*
b.	Lycopsida	(ii)	*Adiantum*
c.	Pteropsida	(iii)	*Psilotum*
d.	Sphenopsida	(iv)	*Equisetum*

Select the **correct** answer from the following

(1) a(iii), b(i), c(iv), d(ii)

(2) a(i), b(iii), c(ii), d(iv)

(3) a(iii), b(i), c(ii), d(iv)

(4) a(i), b(iv), c(iii), d(ii)

128. Chlorophyll bearing simple, thalloid, autotrophic and largely aquatic organisms have been classified into three classes mainly on the basis of

(1) Cell type and cell wall composition

(2) Types of pigments and type of stored food

(3) Cell wall composition and nutrition

(4) Habitat and body organisation

129. In cymose inflorescence

(1) Main axis continues to grow and flowers are borne in acropetal succession

(2) Main axis continues to grow and the flowers are borne in basipetal succession

(3) Growth of main axis is limited and the flowers are borne in basipetal succession

(4) Growth of main axis is limited and flowers are borne in acropetal succession

130. Which of the following plants show whorled and opposite phyllotaxy respectively?

(1) China rose and *Nerium*

(2) *Alstonia* and *Calotropis*

(3) Mustard and Guava

(4) Guava and *Nerium*

131. T.S. of a plant part shows the following features
(i) Ring arrangements of vascular bundles.
(ii) Pericycle consists of semi-lunar shaped patches and is made of sclerenchymatous cells.
Based on the given features, identify the plant part and select the **correct** answer.

(1) Dicot root

(2) Monocot stem

(3) Dicot stem

(4) Monocot root

Vivek Krishna

132. Starch sheath is found in the

(1) Hypodermis of monocot stem

(2) Innermost layer of the cortex in dicot stem

(3) Endodermis of dicot root

(4) Pericycle of dicot stem

133. Match the following **Column-I** with **Column-II** and choose the **correct** option.

	Column-I		Column-II
a.	Imbricate aestivation	(i)	One margin of sepal or petal overlaps that of the next one and so on as in lady's fingers.
b.	Valvate aestivation	(ii)	Margins of sepals or petal overlaps one another but not in any particular direction.
c.	Twisted aestivation	(iii)	The largest petal overlaps two lateral petals which in turn overlap two anterior petals
d.	Vexillary aestivation	(iv)	Sepals or petals in a whorl touch one another at margins without overlapping.

(1) a(ii), b(i), c(iv), d(iii)

(2) a(ii), b(iv), c(i), d(iii)

(3) a(iii), b(i), c(iv), d(ii)

(4) a(iii), b(iv), c(i), d(ii)

134. Stomatal apparatus does **not** include

(1) Guard cells

(2) Subsidiary cells

(3) Mesophyll cells

(4) Stomatal aperture

135. Identify the statements as True **(T)** or False **(F)** w.r.t. *Ulothrix*.

A. It is a colonial alga.
B. Has rigid cell wall made up of an inner layer of cellulose and an outer layer of carrageen.
C. Most common spores produced are the zoospores.
D. Major pigments are chlorophyll *a* and chlorophyll *b*.

(1) A – T, B – F, C – T, D – T

(2) A – F, B – F, C – T, D – T

(3) A – F, B – T, C – F, D – F

(4) A – F, B – T, C – T, D – F

Vivek Krishna

ZOOLOGY

136. The part of a nephron which is not situated in the cortical region of the kidney is

(1) Malpighian corpuscle

(2) PCT

(3) Loop of Henle

(4) DCT

137. Identify the **incorrect** statement w.r.t. human kidney.

(1) They are reddish brown, bean shaped structures.

(2) Each kidney in an adult human measures 10-12 cm in length, 5-7 cm in width and 2-3 cm in thickness.

(3) Average weight of each kidney is about 120-170 gm.

(4) They are situated between the levels of first thoracic and second lumbar vertebra.

138. Complete the following analogy
Cockroach : Malpighian tubules : : Prawn : ____
Choose the **correct** option.

(1) Protonephridia

(2) Green glands

(3) Nephridia

(4) Flame cells

139. Fall in glomerular blood flow/glomerular blood pressure can activate the 'A' to release 'B'.
Choose the **correct** option to fill in the blanks.

(1) **A**-Kupffer cells; **B**-Glomerular filtrate

(2) **A**-Juxta glomerular apparatus; **B**-Rennin

(3) **A**-Juxta glomerular cells; **B**-Renin

(4) **A**-Macula densa; **B**-Angiotensin I

140. Glomerulus along with Bowman's capsule is known as

(1) Malpighian body

(2) Renal tubule

(3) Renal portal system

(4) Malpighian tubule

141. Which one of the following correctly explains the function of the parts of a human nephron?

(1) Proximal Convoluted Tubule – Only 40-50% of the electrolytes are reabsorbed here

(2) Loop of Henle – Maintenance of high osmolarity of medullary interstitial fluid

(3) Collecting duct – Conditional secretion of Na^+ and water

(4) Podocytes – Create minute space for the filtration of blood into the glomerulus from Bowman's capsule

142. Select the **odd** one w.r.t composition of secretions of sweat gland.

(1) NaCl

(2) Small amounts of urea

(3) Lactic acid

(4) Hydrocarbons

143. **Assertion (A):** Nitrogenous waste excreted by land snails is in the form of pellet or paste.
Reason (R): Chief nitrogenous waste excreted by land snails is uric acid that requires minimum water for its elimination.
In the light of above statements, select the **correct** option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

144. Which of the following is absent in dialysing fluid used for hemodialysis in artificial kidney?

(1) Water

(2) Sodium ion

(3) Potassium ion

(4) Urea

145. In human, a comparison of the volume of the filtrate formed per day with that of the urine released suggest that nearly _____ per cent of the filtrate is reabsorbed by the renal tubules.
Fill in the blank by selecting the **correct** option.

(1) 50

(2) 99

(3) 100

(4) 20

146. Path of urine drainage in the human excretory system is

(1) Calyx ⟶ Collecting duct ⟶ Renal pelvis ⟶ Ureter ⟶ Urinary bladder

(2) Renal pelvis ⟶ Collecting duct ⟶ Calyx ⟶ Ureter ⟶ Urinary bladder

(3) Collecting duct ⟶ Calyces ⟶ Renal pelvis ⟶ Ureter ⟶ Urinary bladder

(4) Renal pelvis ⟶ Calyx ⟶ Collecting duct ⟶ Ureter ⟶ Urinary bladder

147. Given below are two statements:
Statement A: In healthy human kidneys, each nephron consists of glomerulus and renal tubule.
Statement B: A number of collecting ducts diverge and open into the calyces through pelvis in the medullary pyramids.
In the light of above statements, choose the **correct** answer from the options given below.

(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Statement A is correct and statement B is incorrect

(4) Statement A is incorrect and statement B is correct

148. Match the column I with column II w.r.t. joints.

	Column I		**Column II**
a.	Ball and socket joint	(i)	Between atlas and axis
b.	Pivot joint	(ii)	Knee joint
c.	Hinge joint	(iii)	Between carpal and metacarpal of thumb
d.	Saddle joint	(iv)	Between humerus and pectoral girdle

Select the **correct** option.

(1) a(iv), b(i), c(ii), d(iii)

(2) a(iv), b(iii), c(i), d(ii)

(3) a(iv), b(ii), c(iii), d(i)

(4) a(ii), b(iv), c(i), d(iii)

149. Zygomatic bone is present in the region of

(1) Ear

(2) Face

(3) Cranium

(4) Larynx

150. Which part of contractile protein has an active ATPase enzyme?

(1) Light meromyosin

(2) Short arm of heavy meromyosin

(3) Globular head of HMM

(4) 'F' actin

Vivek Krishna

151. Select the **incorrect** match w.r.t. number of bones in one hindlimb/forelimb of a man.

(1) Carpals – 8

(2) Phalanges – 14

(3) Metacarpals – 5

(4) Tarsals – 5

152. The total number of bones in each limb of an adult man is

(1) Less than the total number of vertebrae in man

(2) Equal to the total number of ribs in man

(3) More than the sum total of cranial and facial bones in man

(4) Equal to the sum total of phalanges in both forelimbs of man

153. Read the following statements A and B and choose the correct option.
Statement A: Each actin filament is made of two 'F' actins helically wound to each other.
Statement B: Each myosin filament is a polymerised contractile protein.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both statements A and B are incorrect

(4) Both statements A and B are correct

154. Muscle fibres present in the wall of intestine and the wall of heart differ w.r.t

(1) Uninucleated condition

(2) Involuntary contraction

(3) Location of nucleus within the fibres

(4) Shape of the fibres

155. Select the **incorrect** match

(1) *Hydra* – Tentacles

(2) *Paramecium* – Cilia

(3) *Amoeba* – Cilia

(4) *Euglena* – Flagella

156. Select the **correct** statement w.r.t. vertebral column in adult humans.

(1) It is formed by 26 serially arranged units called vertebrae and is ventrally placed in the body.

(2) Each vertebra except atlas has a central canal through which neural canal passes.

(3) First vertebra is the axis and it articulates with the occipital condyles.

(4) The number of cervical vertebrae are seven in almost all mammals including human beings.

157. Which of the following events occur(s) during muscle contraction in skeletal muscle?
(i) 'I' band reduces in length
(ii) 'A' band reduces in length
(iii) 'H' zone greatly reduces in size
Choose the correct option.

(1) (i) and (ii)

(2) (ii) and (iii)

(3) (i) and (iii)

(4) Only (i)

158. In humans, the acromion of scapula articulates with

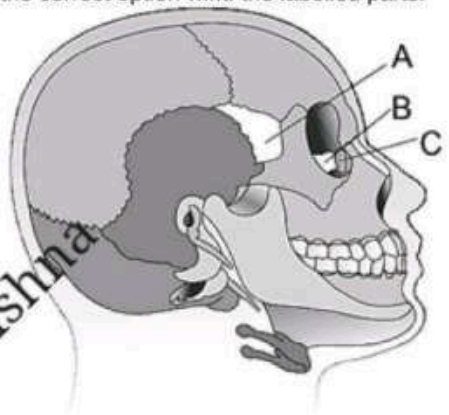
(1) Clavicle

(2) 1st rib

(3) Femur

(4) Humerus

159. Observe the following diagram of a human skull and choose the correct option w.r.t. the labelled parts.



Vivek Krishna

(1) 'A' is a paired cranial bone

(2) 'B' is an unpaired facial bone

(3) Like mandible, C is also an unpaired facial bone

(4) Like the frontal bone, 'A' is also an unpaired cranial bone

160. Which is **correct** sequence of brain meninges from inside to outside?

(1) Arachnoid, dura mater and pia mater

(2) Pia mater, arachnoid and dura mater

(3) Dura mater, arachnoid and pia mater

(4) Pia mater, dura mater and arachnoid

161. Receptors for neurotransmitters are present on the

(1) Pre-synaptic membrane

(2) Post-synaptic membrane

(3) Synaptic cleft

(4) Nodes of Ranvier

162. Choose the part of brain which is not included in brain stem.

(1) Midbrain

(2) Pons

(3) Medulla oblongata

(4) Cerebrum

163. The dorsal portion of the midbrain consists mainly of four round swellings called

(1) Cerebral aqueduct

(2) Corpora quadrigemina

(3) Cerebral hemispheres

(4) Corpus callosum

164. In human brain, the thermoregulatory centre is situated in

(1) Spinal cord

(2) Medulla oblongata

(3) Cerebellum

(4) Hypothalamus

165. Choose the **incorrect** option w.r.t. electrical synapses observed in humans.

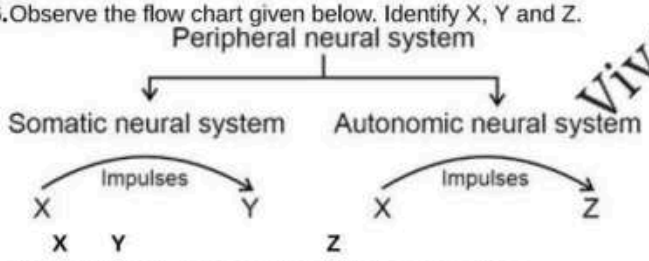
(1) Transmission of an impulse across it is similar to impulse conduction along a single axon

(2) Pre and post-synaptic membranes of neurons are in close proximity

(3) Present abundantly

(4) Neurotransmitters are not involved in the conduction of impulses across it

166. Observe the flow chart given below. Identify X, Y and Z.



	X	Y	Z
(1)	CNS	Involuntary muscles	Voluntary muscles
(2)	PNS	Visceral muscles	Voluntary muscles
(3)	PNS	Cardiac muscles	Unstriated muscles
(4)	CNS	Voluntary muscles	Involuntary organs

(1) (1)

(2) (2)

(3) (3)

(4) (4)

167. The ionic gradients across the resting membrane are maintained by the

(1) Voltage gated ion channels

(2) Sodium-potassium pumps

(3) Electrical synapses

(4) Chemical synapses

168. Which of the following organ systems jointly coordinate and integrate all the activities of the body organs?

(1) Neural system and sensory system

(2) Digestive system and respiratory system

(3) Neural system and endocrine system

(4) Circulatory system and respiratory system

169. Given below are two statements, one is assertion (A) the other is reason (R).

Assertion (A): Cortisol when binds to its receptor does not produce second messengers like cAMP, IP_3, Ca^{++} etc.

Reason (R): Second messengers are generated when hormones interact with membrane bound receptors and cortisol is a steroid hormone which interacts with intracellular receptors.

In the light of above statements, choose the **correct** answer.

(1) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(2) Both (A) and (R) are false

(3) (A) is true, (R) is false

(4) Both (A) and (R) are true and (R) is the correct explanation of (A)

170. Match column I with column II

	Column I		Column II
a.	Emergency hormones	(i)	Insulin
b.	Hypoglycemic hormone	(ii)	Catecholamines
c.	Hypercalcemic hormone	(iii)	Oxytocin
d.	Milk ejecting hormone	(iv)	PTH

Select the **correct** option.

(1) a(i), b(ii), c(iii), d(iv)

(2) a(ii), b(i), c(iv), d(iii)

(3) a(iv), b(iii), c(ii), d(i)

(4) a(ii), b(iv), c(i), d(iii)

171. The steroid hormone that stimulates the formation of alveoli in mammary gland in human females is

(1) Progesterone

(2) Estrogen

(3) Prolactin

(4) Oxytocin

Vivek Krishna

172. In the blood report of a person suffering from simple goitre, increased plasma levels of which of the following hormones is most likely to be reported?

(1) Melatonin

(2) ACTH

(3) TSH

(4) Triiodothyronine

173. In humans, hormone 'X' released from the hypothalamus inhibits the release of GH from anterior pituitary gland.
Identify the hormone representing 'X' and choose the **correct** option.

(1) GnRH

(2) Somatostatin

(3) PRL

(4) FSH

174. Read the following statements.
Statement A: Hormones which interact with nuclear receptors mostly regulate gene expression by the interaction of hormone-receptor complex with the genome.
Statement B: Thyroxine and cortisol are steroidal hormones.
Select the **correct** option.

(1) Only statement A is correct

(2) Both statements A and B are correct

(3) Only statement B is correct

(4) Both statements A and B are incorrect

175. GIP works antagonistically to the hormone which

(1) Is secreted by JG cells of the kidneys

(2) Stimulates the secretion of HCl and pepsinogen

(3) Stimulates the secretion of bicarbonate ions

(4) Acts on gall bladder to release bile juice

176. Which among the following diseases is caused due to hyposecretion of its respective hormones?
(a) Acromegaly
(b) Graves' disease
(c) Addison's disease
(d) Diabetes insipidus
Select the correct option.

(1) (a) and (b)

(2) (c) and (d)

(3) (a), (b) and (c)

(4) (a), (b), (c) and (d)

177. Which of the following pairs of hormones are **not** antagonistic to each other?

(1) Insulin and glucagon w.r.t. blood glucose level

(2) Parathormone and thyrocalcitonin (TCT) w.r.t. blood calcium level

(3) Growth hormone and somatostatin w.r.t. growth of the body

(4) Cortisol and glucagon w.r.t. blood glucose level

178. Which of the following part of brain acts as a coordinating centre for sensory and motor signalling?

(1) Hypothalamus

(2) Hippocampus

(3) Thalamus

(4) Limbic system

179. Which of the following hormones is secreted by an endocrine structure that is situated outside the cranium?

(1) Melatonin

(2) Vasopressin

(3) Thymosin

(4) Oxytocin

180. Which of the following groups of hormones influence the maintenance of water and electrolytes balance in human body?

(1) TCT, ADH, cortisol

(2) Thyroxine, aldosterone, ADH

(3) ADH, glucagon, thymosin

(4) Insulin, cortisol, thymosin

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